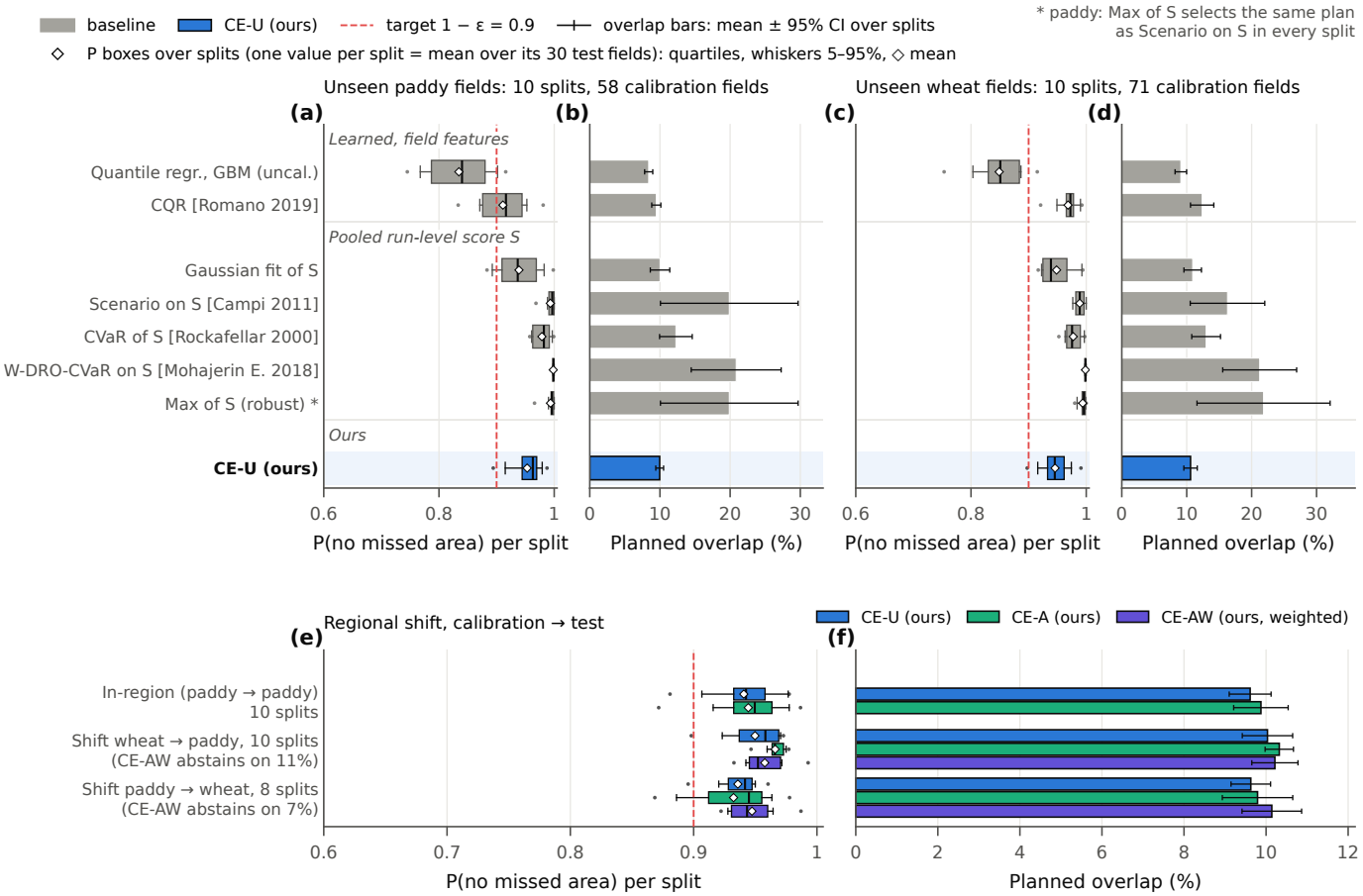
baseline
CE-U (ours)
target 1 − ε = 0.9
overlap bars: mean ± 95% CI over splits
P boxes over splits (one value per split = mean over its 30 test fields): quartiles, whiskers 5–95%, ◇ mean
* paddy: Max of S selects the same plan as Scenario on S in every split
Unseen paddy fields: 10 splits, 58 calibration fields
Unseen wheat fields: 10 splits, 71 calibration fields
(a)
(b)
(c)
(d)
Learned, field features
Quantile regr., GBM (uncal.)
CQR [Romano 2019]
Pooled run-level score S
Gaussian fit of S
Scenario on S [Campi 2011]
CVaR of S [Rockafellar 2000]
W-DRO-CVaR on S [Mohajerin E. 2018]
Max of S (robust) *
Ours
CE-U (ours)
0.6
0.8
1
0
10
20
30
P(no missed area) per split
Planned overlap (%)
Regional shift, calibration → test
(e)
(f)
CE-U (ours)
CE-A (ours)
CE-AW (ours, weighted)
In-region (paddy → paddy)
10 splits
Shift wheat → paddy, 10 splits
(CE-AW abstains on 11%)
Shift paddy → wheat, 8 splits
(CE-AW abstains on 7%)
0.6
0.7
0.8
0.9
1
0
2
4
6
8
10
12
P(no missed area) per split
Planned overlap (%)